EE2005: Electrical Machines
Tutorial 2

1. A 100 kVA, 1100/230 V, 50 Hz transformer has a HV winding resistance of 0.1 Ω and a leakage reactance of 0.4 Ω. The LV winding has a resistance of 0.006 Ω and a leakage reactance of 0.01 Ω. Find the equivalent winding resistance, reactance and impedance referred to the HV and LV sides.

HV: R2/=a^2
LV: R2*=a^2

2. A 230/6 V, single-phase transformer is tested with its secondary winding short circuited. A low voltage (20 V) is applied to the primary terminals and it then takes an input current of 1 A; the power supplied being 10 W. Calculate the values of leakage reactance and resistance, referred to the primary side and then to the secondary side. If the magnetising impedance is neglected, calculate the secondary terminal voltage when a load impedance of value (0.12+j0.09) Ω is connected.

R=P_sc/I^2

3. A 50 kVA, 2400/240 V, 60 Hz distribution transformer has a leakage impedance of (0.72+j0.92) Ω in the high-voltage winding and (0.007+j0.009) Ω in the low-voltage winding. At rated voltage and frequency, the impedance Z_ϕ of the shunt branch (equal to the impedance of R_c and jX_m in parallel) accounting for the exciting circuit is (632+j43.7) Ω when viewed from the low-voltage side.
   The transformer is used to step down the voltage at the load end of a feeder whose impedance is (0.3+j1.6) Ω. The voltage V_s at the sending end of the feeder is 2400 V. Find the voltage at the secondary terminals of the transformer when the load connected to its secondary draws rated current from the transformer and the power factor of the load is 0.8 lagging. Neglect the voltage drops in the transformer and feeder caused by the exciting current.

4. A single-phase load is supplied through a 35 kV feeder whose impedance is (95+j360) Ω and a 35 kV/2400 V transformer whose equivalent impedance is (0.23+j1.27) Ω referred to its low-voltage side. The load is 160 kW at 0.89 leading power factor and 2340 V.

   (a) Compute the voltage at the high-voltage terminals of the transformer.

   (b) Compute the voltage at the sending end of the feeder.

   (c) Compute the power and reactive power input at the sending end of the feeder.

5. A 50 kVA, 2200/110 V transformer when tested gave the following results:
   OC test: measurement on the LV side: 400 W, 10 A, 110 V
   SC test: measurement on the HV side: 808 W, 20.5 A, 90 V
   Compute all the parameters of the equivalent circuit referred to the

   (a) HV side of the transformer

   (b) LV side of the transformer